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Electronic Properties of Metal-/Nitrogen-Doped Graphene Single Atom Catalysts from First Principles

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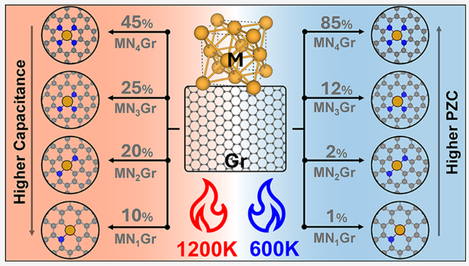
ABSTRACT: Metal-/nitrogen-doped graphene catalysts (MNCs) are a promising class of materials for electrochemical energy conversion, owing to their ability to drive HER and ORR, though their rational design is hindered by the structural heterogeneity produced during synthesis. While traditional computational screening relies on static binding energies, this work utilizes grand-canonical density functional theory (GC-DFT) to rigorously model the potential-dependent electrochemical interface of 232 distinct MNC models. We systematically varied the transition metal center, the number of coordinating nitrogen atoms (MN_xGr, $x = 1-4$), and the local coordination geometry (pyridinic vs pyrrolic) to map out fundamental thermodynamic and electronic properties. Our results reveal a key dichotomy between the coordination geometries: pyridinic sites are consistently more thermodynamically stable, whereas pyrrolic sites exhibit significantly higher interfacial capacitance, indicative of a greater density of states near the Fermi level. Crucially, we demonstrate that thermodynamic stability, as measured by the formation energy, is largely decoupled from key electronic properties like the potential of zero charge (PZC) and capacitance, suggesting these interfacial electrostatic properties can be tuned independently. Furthermore, a Boltzmann analysis demonstrates that higher synthesis temperatures make less stable, under-coordinated structures thermodynamically accessible, providing a theoretical basis for the observed heterogeneity in pyrolyzed materials. This work establishes a foundational map to guide the future rational design of MNCs with tailored functionalities.



■ INTRODUCTION

The development of efficient and robust catalysts is central to addressing global challenges in sustainable energy conversion and chemical production.[1−3] Traditional catalysis heavily relies on expensive and scarce precious metals, such as platinum and iridium, which hinders their large-scale implementation. Electrocatalysis offers a direct pathway to harness renewable electricity for driving thermodynamically demanding chemical transformations, including water splitting,[4] carbon dioxide utilization,[5] and nitrogen fixation.[6] This approach circumvents the need for high temperatures and pressures typical of thermochemical processes and enables decentralized chemical synthesis. However, the sluggish kinetics of many key electrochemical reactions, such as the oxygen reduction reaction (ORR) and CO_2 reduction reaction (CO_2RR), necessitate the design of advanced catalytic materials that can lower activation barriers and enhance reaction rates and selectivity.[7−9]

Among the emerging classes of materials, metal-/nitrogen-doped graphene catalysts (MNCs), a subset of single-atom catalysts (SACs), have garnered significant attention.[10−12] These materials feature isolated metal atoms atomically dispersed and stabilized by nitrogen atoms embedded within a conductive carbon matrix. This unique MN_x coordination environment maximizes atom utilization efficiency and endows the metal center with distinct electronic properties that are highly tunable by modulating the local chemical environment. MNCs thus represent a promising, low-cost alternative to precious metal catalysts, offering a versatile platform for a wide range of electrochemical applications.[13−15]

Over the past decade, extensive research has demonstrated the remarkable potential of MNCs across various critical electrocatalytic processes.[16] In the field of ORR,[17] FeN_xGr materials are now considered the most promising nonprecious metal catalysts for proton-exchange membrane fuel cells and metal-air batteries, with some systems exhibiting activity and durability comparable to the commercial Pt/C benchmark.[18] Advanced characterization techniques, particularly X-ray absorption spectroscopy (XAS), have been instrumental in identifying the planar FeN_4 moiety as the primary active site responsible for this high performance.[19−24]

For the hydrogen evolution reaction (HER), MNCs based on nonprecious metals like Co, Ni, and Mo have been shown to effectively catalyze proton reduction in both acidic and alkaline media.[25,26] The MN_x sites are proposed to modulate the

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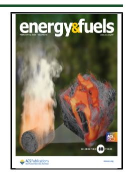

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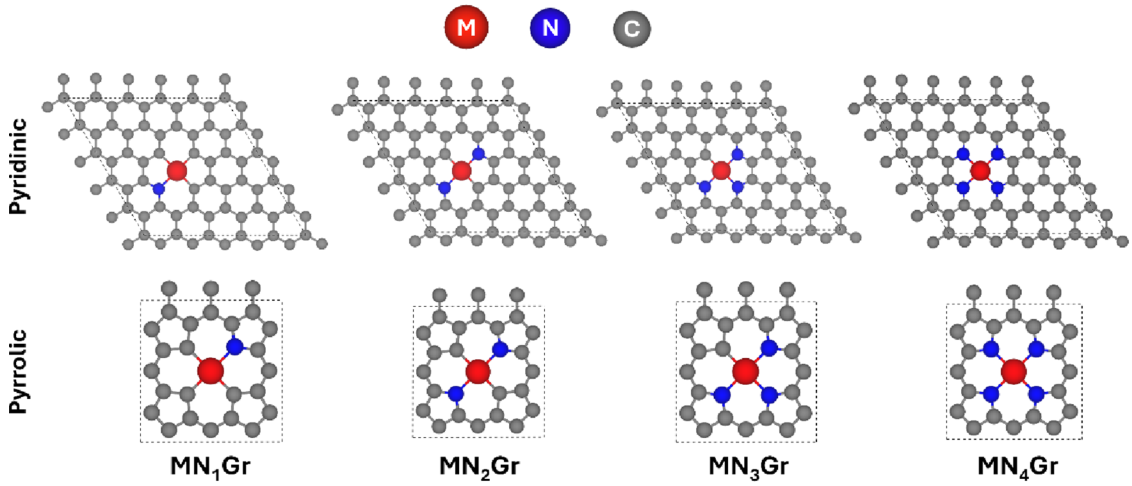

Figure 1. Atomic models of the MN_x active sites under investigation, showing pyridinic (top) and pyrrolic (bottom) coordination. The number of coordinating nitrogen atoms (x) is systematically varied from 1 to 4 for each coordination type. Color scheme: M (red), N (blue), and C (gray).

electronic structure of adjacent carbon atoms and optimize the hydrogen adsorption free energy $\Delta G(H^*)$, thereby lowering the overpotential required for H_2 production. The application of MNCs to the CO_2RR has unveiled a rich landscape of catalytic selectivity.[27−32]

The identity of the metal atom is a key determinant of the catalyst reactivity and product distribution. For instance, NiN_xGr catalysts selectively reduce CO_2 to CO with exceptionally high Faradaic efficiency (>95%).[33−35] On the other hand, Cu-based systems are shown to perform challenging C−C coupling reactions to produce C_{2+} products like ethylene and ethanol.[36,37] The nitrogen reduction reaction (NRR), aiming to convert N_2 to NH_3 under ambient conditions, represents a grand challenge in catalysis.[38] MNCs, particularly those with Fe and Mo active centers inspired by nitrogenase enzymes, have been investigated as potential electrocatalysts to replace the energy-intensive Haber-Bosch process. While reported ammonia yields and Faradaic efficiencies remain low and subject to debate regarding experimental artifacts, these materials offer a viable pathway for exploring a new generation of NRR catalysts.[6,39−41]

The success across these diverse reactions is underpinned by the ability to synthesize materials with a high density of MN_x sites, typically through the pyrolysis of metal salts, nitrogen-containing organic molecules, and carbon supports. Despite significant progress in performance, the materials produced by these methods are often heterogeneous, containing a mixture of MN_x sites with different coordination numbers and geometries, metallic nanoparticles, and various carbon−nitrogen functionalities.[42−44] This structural complexity makes it exceedingly difficult to unambiguously assign catalytic function to a specific active site structure, impeding rational catalyst design.[45−47]

Despite the empirical success of MNCs, a fundamental understanding of the active site's precise atomic structure and its direct relationship to catalytic activity remains incomplete. The MN_4 coordination is widely accepted as a highly active and stable configuration, yet its prevalence and exclusive role as the sole active species in pyrolyzed materials are questionable. The harsh synthesis conditions invariably lead to structural heterogeneity, likely producing a distribution of active sites with varying numbers of coordinating nitrogen atoms. This structural ambiguity obscures the intrinsic reactivity of different MN_x sites and hinders efforts to optimize catalyst synthesis for maximizing the density of the most effective sites.[48−51]

While numerous DFT studies have screened MNCs based on adsorption energies (ΔG) of intermediates, these static calculations often neglect the complex electrochemical environment. Most prior studies employ canonical (constant-charge) DFT, which fails to capture the potential-dependent behavior of the electrode−electrolyte interface. In contrast, this work utilizes Grand Canonical DFT (GC-DFT), often referred to as constant-potential DFT, to treat electrons as an open system at a controlled chemical potential. This methodology allows us to rigorously calculate potential-dependent properties, specifically the potential of zero charge and interfacial capacitance, which are inaccessible to standard computational screening approaches. By applying this formalism to a systematic library of 232 MN_xGr species, we provide a unique data set that tabulates various thermodynamic and electronic properties of two-dimensional doped graphene catalysts, including formation energies, potentials of zero charge, and interfacial capacitance. Specifically, we model MN_x active sites for 29 transition metal atom centers, where the number of coordinating nitrogen atoms, x, is varied from 1 to 4. While certain metals, such as Hg, are unlikely to be stably incorporated into a graphene lattice or have practical catalytic applications, their inclusion allows us to systematically probe trends in electronic properties and thermodynamic stability across the periodic table. As will become evident, such nonfeasible structures deviate significantly from the general trends, thereby reinforcing both the robustness of our computational approach and the reliability of the stability descriptors employed. Furthermore, we explore how these properties are affected by the nature of the nitrogen dopant, shown in Figure 1, comparing the pyridinic configuration (^{i}Gr), where the N atoms are part of a six-membered ring, against the pyrrolic configuration (^{o}Gr), where N atoms are part of a five-membered ring. By calculating the formation energies and key electronic descriptors for this comprehensive set of model structures, we seek to elucidate the thermodynamic stability and intrinsic reactivity of each potential active site. Furthermore, we demonstrate that the stability and electronic properties of MNCs are decoupled and provide insights for the rational design and synthesis of next-generation MNCs with precisely tailored catalytic functions.

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Table 1. Formation Energies (meV/atom) for Metal-/Nitrogen-Doped Graphene Systems, with Metal Centers Coordinated to 1, 2, 3, or 4 Nitrogen Atoms[a]

metal	$MN_1{}^oGr$	$MN_2{}^oGr$	$MN_3{}^oGr$	$MN_4{}^oGr$	$MN_1{}^iGr$	$MN_2{}^iGr$	$MN_3{}^iGr$	$MN_4{}^iGr$
Sc	256	166	71	−22	109	69	32	3
Ti	280	210	134	54	117	86	65	44
V	343	273	199	107	145	115	94	71
Cr	356	277	182	97	153	122	96	62
Mn	318	250	164	97	142	115	85	58
Fe	337	278	197	131	151	126	94	66
Co	347	285	204	140	153	124	90	63
Ni	325	268	190	138	139	111	79	57
Cu	346	306	242	179	142	127	105	84
Zn	365	293	211	132	157	130	103	67
Y	250	157	57	−41	108	68	29	−4
Zr	247	174	99	25	110	77	57	39
Nb	325	262	200	132	141	120	104	91
Mo	376	312	254	170	163	148	136	121
Tc	405	352	245	178	171	153	143	118
Ru	397	334	246	179	170	148	136	110
Rh	360	296	211	145	160	140	113	83
Pd	333	275	197	148	161	132	100	79
Ag	363	322	260	217	170	155	133	127
Cd	379	333	257	180	184	160	131	99
Hf	249	176	100	25	112	80	60	42
Ta	336	274	228	138	145	128	116	107
W	421	348	273	189	178	164	152	136
Re	414	349	262	204	186	170	157	136
Os	395	347	264	190	181	162	149	124
Ir	349	293	211	153	164	146	119	94
Pt	290	242	168	128	145	122	92	73
Au	315	281	226	190	149	138	118	118
Hg	358	347	294	246	176	172	154	115

[a]The pyrrolic and pyridinic phases are denoted by '*o*' and '*i*', respectively.

METHODS

All calculations in this work were conducted using GC-DFT as implemented in the JDFTx software package.[52,53] Electronic structure calculations were performed with the Perdew−Burke−Ernzerhof (PBE) generalized gradient approximation functional, including Grimme's D3 correction to account for van der Waals interactions.[54,55] PBE-D3 was selected to ensure the model remains robust for the material's primary application in electrocatalysis, where capturing van der Waals interactions is a prerequisite for describing surface reactivity. The Kohn−Sham equations were solved using a plane-wave basis with an energy cutoff of 20 hartree, and the Brillouin zone was sampled using a 3 × 3 × 1 Monkhorst−Pack *k*-point mesh. The electronic SCF and geometry optimization were converged to energy thresholds of 1×10^{-7} and 1×10^{-6} hartrees, respectively. To ensure convergence to the correct ground spin state, calculations were spin-polarized, and an initial magnetic moment of 5 μB was applied to the transition metal atom. The pyrrolic MNC supercells contained 25 atoms with in-plane dimensions of 8.41 × 8.41 Å. The pyridinic systems utilize 49 atom supercells with dimensions of 12.51 × 12.30 Å. To prevent interactions across periodic images, a 20 Å vacuum layer was applied along the *z*-axis for both systems. The electrochemical interface was simulated using the CANDLE implicit solvation model, which included an aqueous electrolyte of 0.5 M Na^+ and F^- ions. Because the CANDLE model treats ions as point charges, the results are insensitive to the specific ion identities chosen.[56] The absolute potential of the standard hydrogen electrode (SHE) was determined to be 4.66 V by aligning the computed electrode potential with the experimental work function of the SHE. The computational data for the metal-/nitrogen-doped carbon systems is published and available at the Beyond-DFT Electrochemistry with Accelerated and Solvated Techniques Database (BEAST-DB).[57]

RESULTS

Formation Energies

Formation energies for all MNCs are tabulated in Table 1 and have been calculated using eq 1:

$$\Delta E_f^\circ = E(MN_xC_{y-x}) - E(M) - \frac{x}{2}E(N_2) - (y - x)E(C) \quad (1)$$

where $E(MN_xC_{y-x})$ is the energy of the MNC system, $E(M)$ is the energy per atom of the reference metal in its most stable bulk crystal structure, $E(N_2)$ is the reference energy of nitrogen gas, $E(C)$ is the reference energy per carbon atom in a graphene sheet, x is the number nitrogen atoms to which the metal center is bonded, and y is the total number of nonmetal (carbon and nitrogen) atoms in a unit cell, which is equivalent to 48 and 24 for pyridinic and pyrrolic systems, respectively.

The formation energy, ΔE_f°, is a critical descriptor for evaluating the thermodynamic stability of a given MNC structure.[58−60] It quantifies the energy cost or gain of incorporating an MN_xC_{y-x} defect into a pristine graphene lattice, referenced to the constituent elements: a metal in its bulk ground-state crystal structure, dinitrogen gas, and graphene carbon. A lower or more negative formation energy indicates a thermodynamically more favorable structure that is more likely to be present in synthesized samples, assuming conditions close to thermal equilibrium. Consequently, ΔE_f° serves as a primary screening tool to identify promising and synthesizable catalyst

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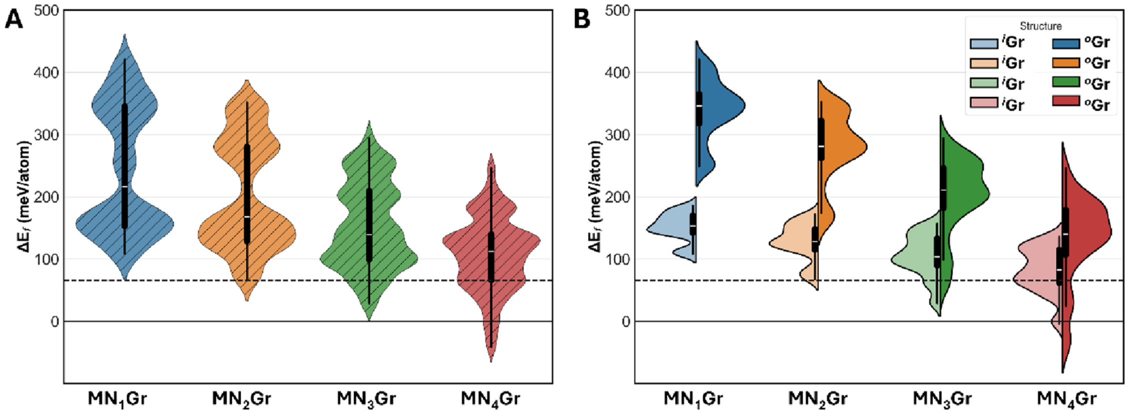

Figure 2. Formation energy distributions of MN_xGr systems, sorted by the number of nitrogen atoms coordinated to the metal center. The horizontal dashed line corresponds to the formation energy of $FeN_4{}^iGr$ at 66 meV/atom. Panel A shows pyridinic and pyrrolic configurations in a combined data set, whereas panel B differentiates between the two data sets, with the pyridinic configurations plotted in a lighter shade. Two distinct trends are observed: first, structural stability correlates with the number of coordinated nitrogen atoms, and second, pyridinic configurations demonstrate greater stability than their pyrrolic counterparts.

structures and to understand their resistance to degradation via atom leaching or structural rearrangement.

As shown in Table 1, the calculated formation energies for the majority of the MNC structures are positive. This is expected, as these structures represent defects within the exceptionally stable aromatic lattice of graphene. The creation of such defects is an endothermic process relative to the pristine sheet and bulk elemental precursors. However, positive formation energies do not preclude their existence. MNCs are typically synthesized via high-temperature pyrolysis, a nonequilibrium process where kinetic factors dominate. At elevated temperatures, sufficient energy is available to overcome activation barriers, allowing for the formation of these thermodynamically metastable structures. Upon cooling, these defects become kinetically trapped within the carbon matrix. Therefore, MNCs with moderately positive formation energies are considered synthesizable and can persist as active catalytic sites under operating conditions. This is empirically supported by the successful fabrication of single-atom catalysts using metals beyond the standard iron and nickel systems, such as molybdenum and ruthenium. Specifically, recent works have demonstrated the synthesis of MoN_4Gr,[61] RuN_4Gr,[62] and PdN_4Gr, confirming that these metastable structures are experimentally accessible.

A comprehensive analysis of the formation energies reveals several key trends regarding the stability of MNC sites. The most dominant trend, illustrated in the overall distributions in Figure 2A, is the inverse relationship between formation energy and the nitrogen coordination number. For nearly all metals considered, the stability of the active site increases as the number of coordinating nitrogen atoms increases. The median formation energy systematically decreases from MN_1Gr to MN_4Gr structures, indicating a strong thermodynamic driving force for the metal center to be coordinated by a greater number of nitrogen atoms to promote the formation of more stable, and often more reactive, MN_3Gr and MN_4Gr sites.

The second critical factor governing stability is the coordination geometry, namely pyridinic versus pyrrolic. A systematic comparison of the data in Table 1 shows unequivocally that pyridinic coordination imparts significantly greater stability than pyrrolic coordination. For any given metal and nitrogen coordination number x, the formation energy of the pyridinic configuration is consistently lower, often by 100 meV/atom or more. This pronounced stability difference is clearly visualized in the bimodal distributions of Figure 2B. For each coordination number, the distribution splits into two distinct clusters; the lower-energy cluster corresponds to the pyridinic structures, and the higher-energy cluster corresponds to the less stable pyrrolic structures. This indicates that the formation of a six-membered pyridinic ring provides a more effective environment for stabilizing the embedded metal atom compared to the five-membered ring structure.

Another significant trend revealed in Figure 2B is the relationship between the nitrogen coordination number, x, and the relative stability of the two coordination geometries. The energetic separation between the lower-energy pyridinic distribution and the higher-energy pyrrolic distribution systematically decreases as x increases from 1 to 4. For MN_1Gr structures, the stability gap between the two configurations is very large, indicating a strong thermodynamic penalty for forming a pyrrolic site. This is confirmed by the data in Table 1, where for a typical metal like Fe, the pyrrolic configuration is 186 meV/atom less stable than the pyridinic one.

As the number of coordinating nitrogens increases, this energy difference consistently shrinks. For Fe, the stability gap reduces to 152 meV/atom for $x = 2$, 103 meV/atom for $x = 3$, and just 65 meV/atom for $x = 4$. This suggests that as the metal's first coordination sphere becomes saturated with nitrogen atoms, the type of nitrogen coordination becomes less critical to the overall stability of the complex. The local MN_4 moiety itself becomes the dominant factor in stabilizing the structure, reducing the thermodynamic penalty associated with the less favorable pyrrolic carbon framework. This implies that while pyridinic sites are nearly always the preferred geometry, the formation of highly coordinated pyrrolic sites ($MN_3{}^oGr$ and $MN_4{}^oGr$) may become more competitive in nitrogen-rich synthesis conditions.

Thermal Populations

While formation energies identify the most thermodynamically stable structure at 0 K, the actual distribution of MNC sites in a synthesized material is dictated by the high temperatures used during fabrication processes like pyrolysis.[63,64] At elevated temperatures, the system possesses sufficient thermal energy k_BT to overcome energy barriers and populate not only the lowest-energy ground state but also higher-energy, less stable configurations. To model this thermodynamic reality, we can

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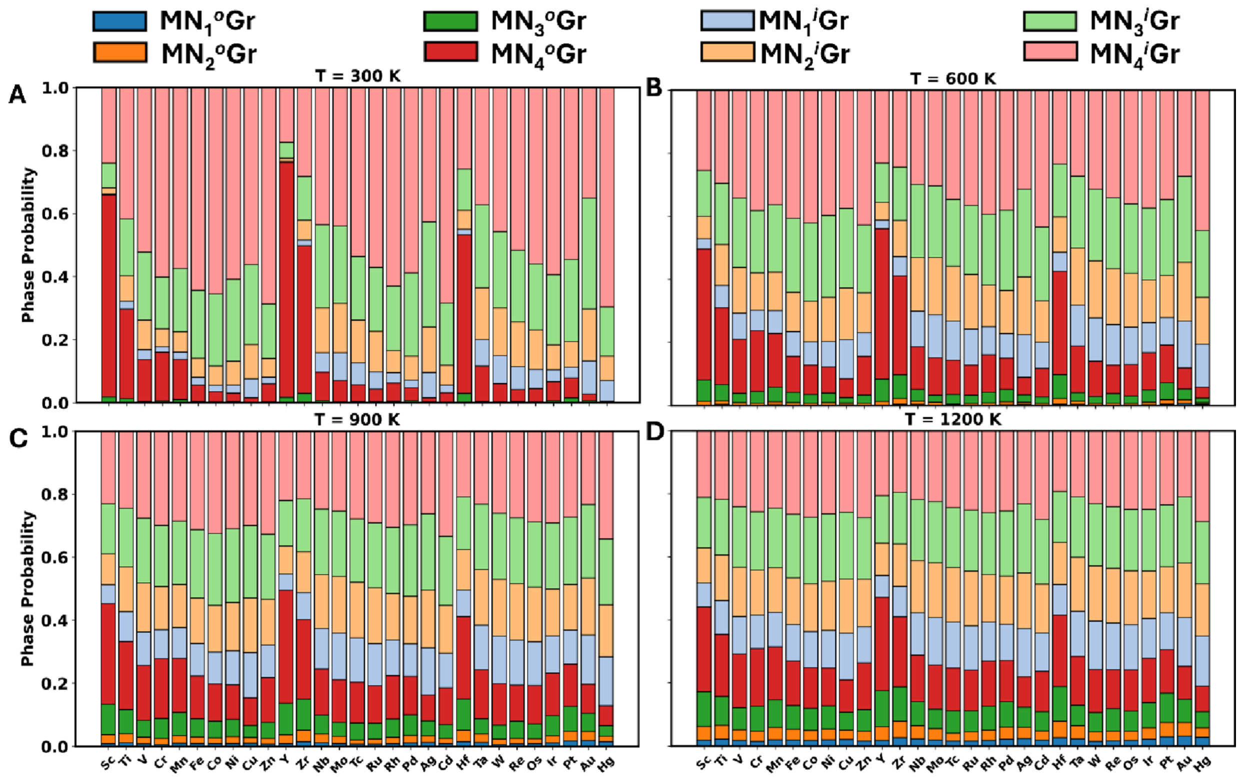

Figure 3. Thermal populations of the 8 different phases of each metal-/nitrogen-doped graphene at 300 K (A), 600 K (B), 900 K (C), and 1200 K (D), based on Boltzmann statistics of their formation energies. Color scheme: MN_4Gr (red), MN_3Gr (green), MN_2Gr (orange), and MN_1Gr (blue). Pyridinic structures are depicted in lighter shades of the same colors.

use the Boltzmann distribution to calculate the probability of forming each of the eight distinct MNC structures at a given synthesis temperature.

The probability, P_i, of finding a metal atom in a specific coordination environment i, is given by the Boltzmann equation[65]:

$$P_i = \frac{e^{-E_i/k_B T}}{Z} \tag{2}$$

Here, E_i is the formation energy of state i, taken from Table 1, k_B is the Boltzmann constant, and T is the synthesis temperature in Kelvin. The denominator, Z, is the partition function, which is the sum of the Boltzmann factors of all N possible states:

$$Z = \sum_{i=1}^{N} e^{-E_i/k_B T} \tag{3}$$

In our work, N is equal to 8 for each metal as we consider the four pyridinic and four pyrrolic structures. A more in-depth study would investigate all possible deformations. Z normalizes the probabilities, ensuring they sum to 1. This equation shows that while low-energy states are always favored, their dominance diminishes as temperature increases. The term $\frac{E_i}{k_B T}$ shows that the energy penalty for occupying a less stable state is effectively softened by higher thermal energy, making higher-energy structures more statistically accessible.

Figure 3 visualizes these calculated phase probabilities for each metal at four distinct temperatures ranging from 300 to 1200 K, providing a map of the expected structural distribution. At 300 K as seen in panel A, systems overwhelmingly occupy the most stable state, or states assuming they exhibit nearly equivalent formation energies. For most metals, that is nearly the pyridinic $MN_4{}^iGr$ (light red) and $MN_3{}^iGr$ (light green) states. Four outliers (Sc, Y, Zr, and Hf) prefer the pyrrolic $MN_4{}^oGr$ state (dark red) at low temperatures due to lower formation energies relative to their pyridinic counterparts. As the temperature increases through panels B, C, and D, a clear trend emerges. The probability distribution broadens significantly. The dominance of the single most stable state decreases, and the probabilities of higher energy, less stable structures increase. Less stable configurations, like MN_2Gr (orange) and MN_1Gr (blue), become more accessible and are expected to be present in the final material. The bars corresponding to these structures, which are negligible at 300 K, grow significantly at 900 and 1200 K. This finding provides a fundamental explanation for the structural heterogeneity observed in experimentally synthesized MNC catalysts. The model suggests that the relative populations of different active sites can be intentionally tuned by controlling the synthesis temperature, offering a powerful strategy for catalyst design.

These observations are supported by experimental findings which indicate that high-temperature synthesis can favor pyrrolic configurations under specific conditions. Lai et al. demonstrated that pyridinic nitrogen is thermally unstable at annealing temperatures as high as 1000 °C, leading to a significant decrease in pyridinic content and a concurrent rise in pyrrolic species.[66] Similarly, Menga et al. reported that pyrrolic FeN_4Gr sites can remain the dominant active species even after heat treatment at 1000 °C.[67] Consequently, our results rely on the assumption that catalytic graphitization, a process by which metal atoms lower the activation barrier for reorganizing amorphous carbon into highly ordered graphite, is effectively suppressed. By inhibiting lattice reorganization, the necessary

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defects for pyrrolic coordination are preserved even at elevated synthesis temperatures.

Potentials of Zero Charge

Beyond thermodynamic stability, the potential of zero charge is a critical electronic property that governs the catalyst's behavior at the electrochemical interface. The PZC is the specific electrode potential at which the net charge on the catalyst's surface is zero. In this work, the PZC is defined as the Fermi level of the neutral surface referenced to the Standard Hydrogen Electrode (SHE). To convert the vacuum-referenced Fermi level to this electrochemical scale, the absolute potential of the SHE (V_{ref}) is subtracted, as presented in eq 4:

$$\mathrm{PZC(vs\ SHE)} = E_{\mathrm{fermi}} - V_{\mathrm{ref}} \tag{4}$$

For the CANDLE solvation model implemented in this work, the calibrated V_{ref} is 4.66 V. The PZC serves as an intrinsic measure of the surface's work function in an aqueous environment, indicating how readily the catalyst holds onto or gives up electrons. Figure 4 plots the PZC for all 29 metals across

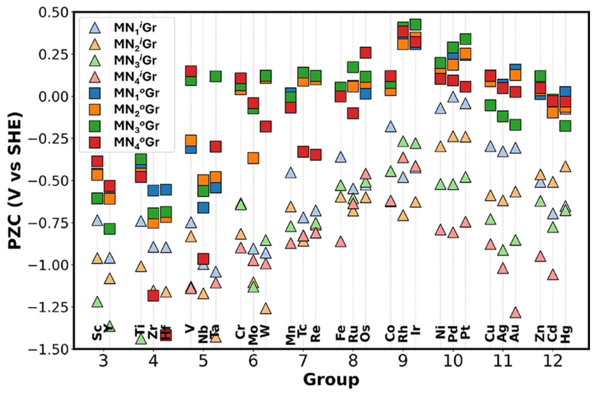

Figure 4. Potential of zero charge (PZC) values for all 29 MNC systems and each of their 8 configurations. Pyrrolic systems are plotted as bright square data points, whereas their pyridinic counterparts are plotted as faint triangles, with matching colors representing each of the four nitrogen-coordination environments.

their eight structural configurations versus their metal group number, revealing systematic trends. The PZC for every MNC configuration is tabulated in Table 2.

A striking trend immediately visible in the data is the consistent offset between pyrrolic and pyridinic structures. For any given metal and nitrogen coordination number, the pyrrolic configuration always exhibits a more positive PZC than its pyridinic counterpart. A higher PZC indicates that a surface is intrinsically more negatively charged and by definition, neutralized at this high positive potential. Conversely, a more negative PZC signifies an electron-deficient surface with a lower work function, as it requires highly reducing potentials to keep the surface neutral.

The PZC is a fundamental descriptor that can help predict a catalyst's suitability for different classes of electrochemical reactions.[24,68,69] The charge of the catalyst's surface is determined by the applied potential U relative to its PZC: The surface is negatively charged when U < PZC and positively charged when U > PZC. This relationship allows us to infer catalytic tendencies of different MNCs. Cathodic reactions, such as CO_2RR and HER, require the electrocatalyst to facilitate electron transfer from the external potential source to a reactant. The facilitation of such electron transfer necessitates a negatively charged catalyst surface. The energetic efficiency of achieving this state is governed by the catalyst's PZC. For optimal performance, the catalyst's PZC should be slightly more positive than the desired reaction potential U. This configuration ensures the surface becomes negatively charged at the operating potential while minimizing the required overpotential. Materials with a PZC in this optimal range, such as pyridinic sites with late transition metals and pyrrolic sites with early transition metals, are therefore considered favorable for these processes.

On the other hand, anodic reactions, such as ORR, involve the transfer of electrons from a reactant to the catalyst. Such processes are favored on positively charged surfaces. A surface becomes positively charged when the applied potential is more positive than its PZC. Since anodic reactions are carried out at more positive potentials, the most efficient catalyst will have a PZC that is slightly less positive than the desired reaction potential. This ensures the surface achieves the necessary positive charge with the smallest possible energy input, making materials with a high PZC generally suitable for these applications. Consequently, pyrrolic sites, may effectively promote the adsorption of anionic intermediates like hydroxide.

Crucially, this guideline is not set in stone. As Figure 4 clearly shows, the PZC is a highly tunable property, and both the pyridinic and pyrrolic families span a wide range of potential values depending on the specific choice of the central metal and the nitrogen coordination number. More fundamentally, while the PZC is a critical parameter for governing the initial electrostatic interactions that influence reactant adsorption, this is not the sole determinant of catalytic activity. The overall performance of an electrocatalyst is ultimately dictated by the complete energetic landscape of the reaction pathway, which encompasses all elementary steps from initial reactant adsorption and activation to the final desorption of products.

Oxidation States

To further understand the electronic properties of the MNC active sites, we analyzed the oxidation states of the central metal atom using computational charge-partitioning schemes. The calculated oxidation state is a powerful descriptor that relates directly to the metal center's ability to engage in redox chemistry.[59,70,71]

Figure 5A shows the metal's oxidation state, calculated using the Bader formalism, as a function of its group in the periodic table. The data reveals a distinct V-shaped trend. Early transition metals in Groups 3−5, like Sc, Zr, and V, are highly electropositive and readily donate electron density to the N_xGr framework, resulting in high positive oxidation states, often greater +2.0. Later transition metals in Groups 8−10, such as Fe, Co, and Pd, are more electronegative. Their tendency to donate electrons decreases, leading to a minimum in the oxidation state, typically between +0.5 and +1.0. The trend then rises again for the post-transition metals of Groups 11 and 12. This rise is likely due to the stable, filled *d*-electron shells of Group 11 and 12 metals, which promotes the loss of their outer *s* electrons and results in a higher formal positive charge. A distinct periodic trend is observed for the early transition metals, where the first row 3d transition metals consistently exhibit lower Bader oxidation states than their heavier 4d and 5d analogs. This trend reaches a minimum and eventually reverses from approximately Group 8 onward.

A comparison between the Bader[72,73] and Löwdin[74] charge analysis methods is presented in the parity plot in Figure 5B. While both methods show a strong linear correlation, indicating

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Table 2. Potential of Zero Charge (PZC) Values (mV vs SHE) in Water for All Metal-/Nitrogen-Doped Graphene Systems Modeled in This Work

metal	$MN_1{}^{o}Gr$	$MN_2{}^{o}Gr$	$MN_3{}^{o}Gr$	$MN_4{}^{o}Gr$	$MN_1{}^{i}Gr$	$MN_2{}^{i}Gr$	$MN_3{}^{i}Gr$	$MN_4{}^{i}Gr$
Sc	−461	−565	−623	−693	−736	−958	−1218	−1609
Ti	−562	−481	−457	−490	−740	−1006	−1438	−1530
V	−341	−291	52	108	−686	−847	−1147	−1135
Cr	80	34	9	−132	−619	−822	−631	−897
Mn	22	−32	−72	−79	−450	−654	−746	−870
Fe	79	1	15	1	−348	−567	−592	−863
Co	85	29	252	126	−186	−365	−436	−620
Ni	207	160	72	74	−62	−314	−517	−790
Cu	111	−48	−88	48	−297	−587	−731	−878
Zn	27	58	17	6	−504	−468	−621	−948
Y	−605	−699	−799	−822	−957	−1081	−1364	−1709
Zr	−804	−757	−816	−1232	−893	−1152	−1665	−1668
Nb	−643	−580	−665	−1000	−995	−1170	−1578	−1600
Mo	−394	−416	−50	−57	−903	−1087	−1130	−974
Tc	121	116	153	−310	−725	−855	−833	−826
Ru	92	90	175	−217	−549	−656	−608	−636
Rh	361	345	404	148	−479	−707	−259	−366
Pd	267	185	68	49	−6	−237	−522	−807
Ag	85	−117	−154	−60	−326	−618	−912	−1016
Cd	−342	−186	−62	−139	−695	−506	−775	−1056
Hf	−782	−723	−797	−1306	−893	−1160	−1653	−1711
Ta	−537	−514	111	−306	−1040	−1423	−1552	−1105
W	179	131	105	−194	−928	−1266	−854	−992
Re	135	116	120	−226	−664	−787	−743	−809
Os	10	99	109	253	−524	−599	−515	−457
Ir	403	373	408	69	−425	−625	−278	−411
Pt	286	238	65	7	−42	−240	−478	−744
Au	142	−89	−163	−246	−306	−566	−852	−1281
Hg	−332	−177	−213	−81	−652	−417	−676	−75

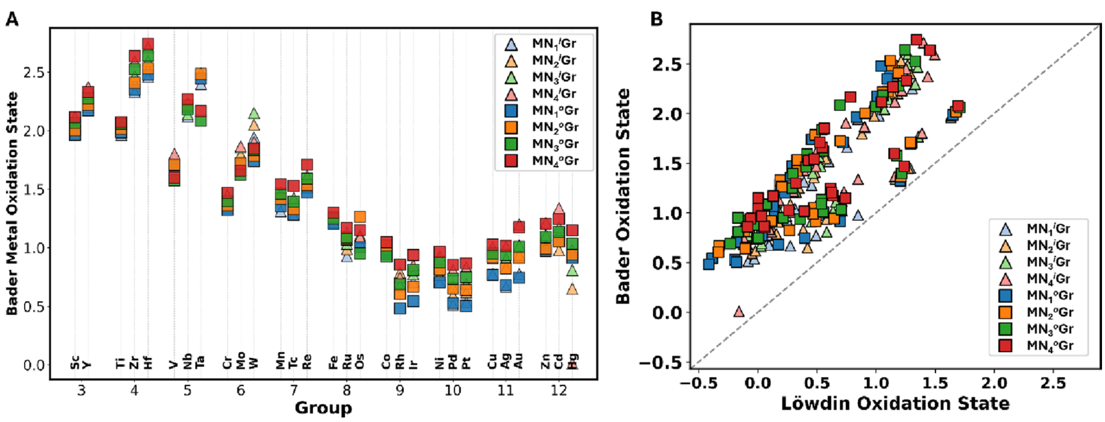

Figure 5. (A) Oxidation states of the metal centers in MN_xGr as a function of group number on the periodic table. (B) Parity plot of oxidation states as determined by Bader and Lowdin charge density analyses. Pyrrolic systems are represented by bright squares, whereas pyridinic systems are plotted as fainter triangles, with colors matching their respective nitrogen coordination environment.

they capture the same chemical trends, the Löwdin method consistently calculates a lower oxidation state than the Bader method by approximately 0.5 to 1.0 charge units, as all data points lie above the $y = x$ parity line. This systematic difference stems for the fact that Löwdin charge partitions charge based on atomic orbitals, and in doing so, it equally divides the electrons in a chemical bond between the two atoms. In a polar M−N bond, this artificially assigns more electron density to the electropositive metal than it would realistically possess, thereby underestimating its positive charge. On the other hand, Bader charge partitions the system's electron density in real space based on zero-flux surfaces, providing a more physically intuitive measure of charge separation.

An important observation from Figure 5A is that for any given metal, the calculated Bader oxidation states are tightly clustered together, regardless of the nitrogen coordination number or the specific geometry. This strongly suggests that the oxidation state of the active site is overwhelmingly dictated by the intrinsic chemical identity of the metal itself. The local coordination environment acts only as a minor perturbation, while the element's electronegativity and electronic structure are the primary factors controlling the charge at the catalytic center.

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Capacitance

In addition to thermodynamic and electronic properties, the interfacial capacitance provides critical insight into the behavior of the MNC materials under an applied potential.[75,76] The total capacitance measures the ability of the electrode−electrolyte interface to store charge. Here, capacitance is computed directly as the differential response of the electron count to the applied bias, normalized per surface area, as defined in eq 5:

$$C = \frac{\Delta n_{\mathrm{elec}}}{\Delta V \times A} \tag{5}$$

While capacitance is often estimated in canonical DFT by integrating the Density of States (DOS) near the Fermi level posthoc, the GC-DFT scheme employed in this work treats the electron number as a dynamic variable that equilibrates with the applied potential during the SCF cycle. This allows for a direct calculation of the capacitance as a thermodynamic observable under operating conditions, rather than relying on static electronic structure approximation. This explicit optimization of the electron count renders posthoc DOS integration redundant, as the differential capacitance inherently accounts for orbital relaxation effects that static DOS approximations neglect. Accordingly, we calculated capacitance over the cathodic potential window of 0 to −1 V vs SHE. In this context, the computed capacitance serves as a robust proxy for the Electrochemical Active Surface Area (ECSA) and reflects, where a higher value suggests that more active sites are electronically accessible under bias and that the material has a greater capacity for the charge transfer processes that drive catalysis.[77−79]

Figure 6 plots the total capacitance for all MNC systems, revealing several key trends. The most dominant and consistent trend is the remarkable difference between the capacitance of pyridinic and pyrrolic geometries. Across the entire range of metals and coordination numbers, the pyrrolic structures consistently exhibit a significantly higher capacitance than their pyridinic counterparts. For many metals, the capacitance of the pyrrolic sites is double or even triple that of the corresponding pyridinic sites.

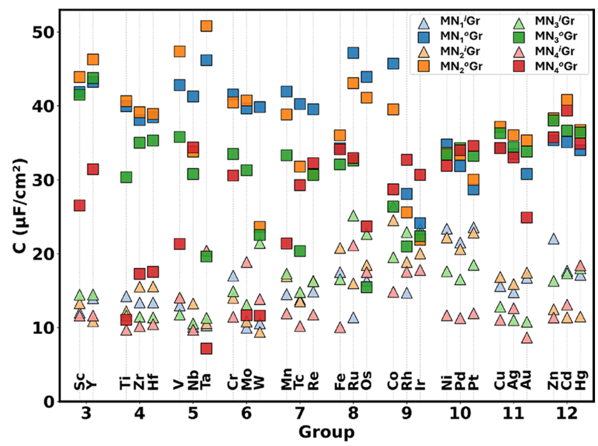

Figure 6. Capacitance (μF/cm^2) of metal-/nitrogen-doped graphene systems, calculated over an applied potential swing of 1 V, ranging from 0 to −1 V vs SHE. Pyrrolic systems are represented by bright squares, whereas pyridinic systems are plotted as fainter triangles, with colors matching their respective nitrogen coordination environment.

This observation has a profound implication for the electronic structure of the active sites. A higher capacitance indicates a greater ability to accommodate charge, which directly points to the pyrrolic systems possessing a higher DOS near the Fermi level. When the potential is applied, a material with a higher DOS has more available electronic states to fill or empty, resulting in greater charge storage and thus a higher measured capacitance. This suggests that the five-membered ring structure of the pyrrolic configuration induces an electronic environment at the metal center that is fundamentally more metallic or has more states available for redox processes compared to the pyridinic configuration.

While the coordination geometry is the primary determinant, other factors also influence capacitance. The choice of metal leads to significant variation, although the trend across the periodic table is not as simple as those for stability or oxidation state. Early transition metals tend to show very high capacitance in the pyrrolic form. In contrast, the effect of the nitrogen coordination number appears to be a secondary factor, causing smaller variations compared to the dramatic split between the pyridinic and pyrrolic families. This analysis shows that the coordination geometry not only impacts thermodynamic stability and the potential of zero charge but also fundamentally alters the electronic structure at the Fermi level. The enhanced DOS of pyrrolic sites, evidenced by their higher capacitance, may make them particularly interesting for catalytic reactions that are sensitive to the availability of electronic states at the active site.[51,80]

Structure−Property Trends

To reveal the underlying relationships between the calculated physical properties, key descriptors were plotted against one another as shown in Figure 7. Figure 7A reveals a strong inverse correlation between the calculated Bader oxidation state of the metal center and the PZC of the MNC. This relationship is expected, as both descriptors probe the underlying electron density and effective charge of the active site. Contrary to the behavior expected for simple bulk metals, the observed inverse trend highlights the dominant role of charge transfer between the metal and the N-doped graphene support. A more positive oxidation state on the metal center signifies a greater donation of its electron density to the surrounding nitrogen−carbon moiety. While this leaves the metal atom itself electron-poor, it renders the surrounding N_xGr environment electron-rich. This electron-rich N_xGr environment results in a more negative PZC, and hence, a lower work function, as the electrons are dispersed throughout the graphene sheet and not localized on the metal center. Furthermore, a consistent offset is observed between the two coordination families. For any given metal oxidation state, the pyridinic environment systematically results in a more negative PZC compared to the pyrrolic configuration. This suggests that the specific coordination geometry induces a distinct local surface dipole that further modulates the PZC.

Conversely, the relationship between the formation energy ΔE_f° and PZC, shown in Figure 7B, exhibits no clear correlation. The wide scattering of data points indicates that these two properties are largely decoupled. This decoupling arises because the two descriptors are governed by distinct physical drivers following different periodic trends. The PZC is dependent primarily on the identity of the transition metal and its intrinsic electronic properties, such as the ionization energy and electron affinity. On the other hand, thermodynamic stability is dominated by geometric factors, such as the ionic radius and the efficiency of orbital hybridization between the metal center and its coordination environment. This finding is significant, as

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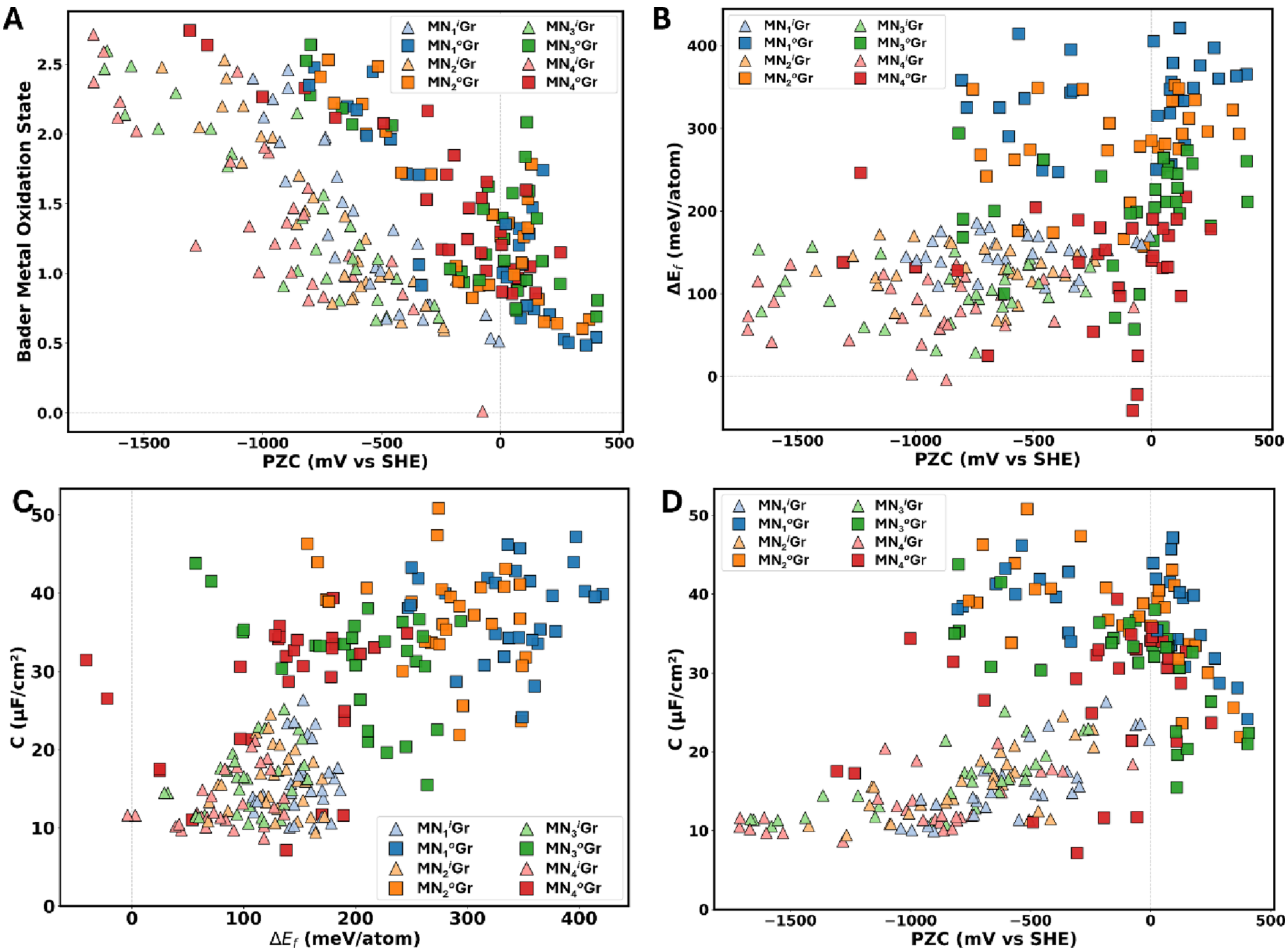

Figure 7. (A) Metal center Bader oxidation state versus PZC (mV vs SHE). (B) Formation energies (meV/atom) versus PZC (mV vs SHE). (C) Capacitance (μF/cm^2) versus formation energies (meV/atom). (D) Capacitance (μF/cm^2) versus PZC (mV vs SHE). Pyrrolic systems are represented by bright squares, whereas pyridinic systems are plotted as fainter triangles, with colors matching their respective nitrogen coordination environment.

it suggests that the thermodynamic stability of a defect site does not dictate its electronic work function. This implies that these properties can be engineered independently to optimize a catalyst for both stability and electronic activity.

The relationship between capacitance and formation energy is shown in Figure 7C. A clear segregation between the two coordination geometries is observed. The pyridinic structures are confined to a low-capacitance region, generally less than 25 μF/cm^2, while the pyrrolic structures span a much wider range, reaching values above 40 μF/cm^2. As with the PZC, capacitance shows no direct correlation with formation energy, reinforcing the concept that the electronic and thermodynamic properties are not directly linked.

Finally, Figure 7D, which plots capacitance versus PZC, serves as an electronic property map. Although no universal correlation exists, the plot delineates the accessible property space for each configuration. The pyridinic structures are confined to the low-capacitance region, whereas the pyrrolic structures populate the high-capacitance region across a wide range of PZC values. This demonstrates that if a high capacitance is a desired feature for a catalyst, the pyrrolic coordination geometry is the more promising synthetic target.

Collectively, this systematic analysis of the relationships between key physical descriptors deconstructs the complex interplay between structure and electronic properties in MNC systems. The distinct and often decoupled roles of metal identity, nitrogen coordination, and local geometry have been elucidated. These established structure−property relationships provide a foundational framework to guide the rational design of catalysts with precisely engineered functionalities.

Implications for Rational Catalyst Design

By mapping the fundamental descriptors calculated herein to the catalytic applications introduced earlier, we can propose specific design guidelines for targeted reactions. Electrochemical reduction necessitates surfaces that facilitate electron transfer to adsorbed reactants. This mechanism is electrostatically favored on surfaces with a PZC positive relative to the operating potential, ensuring the surface maintains a net negative charge during operation. Data presented in Figure 4 indicates that pyridinic-coordinated late transition metals, specifically the Group 10 elements Ni, Pd, and Pt, exhibit PZCs well-aligned with the potentials required for HER and CO_2RR. These findings provide a theoretical basis for the experimental success of NiN_4Gr catalysis in CO_2 reduction.

The ORR involves oxygenated intermediates and proton-coupled steps that require efficient reactant access. Since the reaction operates at highly positive potentials (0.8−0.9 V vs

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SHE),[19] it is advantageous to select a catalyst with a PZC slightly more positive than this range. This ensures the surface remains neutral or slightly negatively charged during operation, minimizing the accumulation of positive charge. A highly positive surface is detrimental as it repels protons and overstabilizes anionic species, which can block active sites, thereby increasing the kinetic barrier. Consequently, minimizing the deviation between the operating potential and the catalyst's PZC serves as a universal design heuristic to reduce parasitic interfacial field effects and avoid the energy waste associated with excessive double-layer charging.

Local electronic structure serves as a second fundamental criterion that complements the PZC. The active metal center must exhibit an oxidation state within a specific intermediate range to balance the reaction. This intermediate cationic character ensures the site possesses sufficient electrophilicity to activate molecular oxygen while avoiding the excessive positive charge that leads to product inhibition. The established ORR benchmark, FeN_4Gr in its pyridinic structure, validates this design framework, as it combines the required PZC alignment with an intermediate oxidation state of +1 for reversible intermediate binding.

While thermodynamic stability favors pyridinic sites, Figure 6 shows that pyrrolic sites exhibit significantly higher capacitance. This higher density of states at the Fermi level suggests that pyrrolic sites may act as superior electron reservoirs. This property could be crucial for driving kinetically complex, multielectron transfers, such as the reduction of CO_2 to C_{2+} products, where maintaining a high population of surface electrons is necessary to stabilize key intermediates, although difficult to control due to synthesis and stability limitations.

■ CONCLUSIONS

In conclusion, we have performed a systematic, first-principles investigation into the fundamental thermodynamic and electronic properties of 232 distinct metal-/nitrogen-doped graphene catalysts. Our analysis of formation energies, potentials of zero charge, Bader oxidation states, and interfacial capacitance has established clear structure−property relationships. The results reveal a fundamental disparity: pyridinic MNCs are universally more thermodynamically stable, while pyrrolic sites consistently exhibit higher capacitance, indicative of a greater density of electronic states near the Fermi level. Critically, we find that the thermodynamic stability and key electronic properties are largely decoupled, suggesting that they can be independently optimized through the strategic selection of the metal, nitrogen coordination, and local geometry.

While the present work establishes a comprehensive map of intrinsic material properties, the ultimate goal is to connect these properties to catalytic function. The well-defined structural models presented herein provide a robust platform for future computational investigations into explicit reaction mechanisms under applied electrochemical potentials. By calculating the binding energies of key reaction intermediates (*COOH, *OOH, *H, etc.) and determining the activation barriers for elementary steps, it is possible to construct full catalytic cycles for various MNC configurations. Such mechanistic studies would directly reveal how the intrinsic descriptors identified in this work, such as the PZC and metal oxidation state, govern catalytic activity and selectivity. This provides a direct pathway for the rational screening of catalyst candidates for targeted electrochemical systems, such as CO_2RR or OER.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Wang, J.; Zhang, Y.; Ma, Y.; Yin, J.; Wang, Y.; Fan, Z. Electrocatalytic Reduction of Carbon Dioxide to High-Value Multicarbon Products with Metal−Organic Frameworks and Their Derived Materials. *ACS Materials Letters* **2022**, *4*, 2058−2079.

(2) Kanan, M. W.; Nocera, D. G. In Situ Formation of an Oxygen-Evolving Catalyst in Neutral Water Containing Phosphate and CO_{2+}. *science* **2008**, *321*, 1072−1075.

(3) Huber, G. W.; Iborra, S.; Corma, A. Synthesis of Transportation Fuels from Biomass: Chemistry, Catalysts, and Engineering. *Chem. Rev.* **2006**, *106*, 4044−4098.

(4) Jaramillo, T. F.; Jørgensen, K. P.; Bonde, J.; Nielsen, J. H.; Horch, S.; Chorkendorff, I. Identification of Active Edge Sites for Electrochemical H_2 Evolution from MoS_2 Nanocatalysts. *Science* **2007**, *317*, 100−102.

(5) Nitopi, S.; Bertheussen, E.; Scott, S. B.; Liu, X.; Engstfeld, A. K.; Horch, S.; Seger, B.; Stephens, I. E.; Chan, K.; Hahn, C. Progress and Perspectives of Electrochemical CO_2 Reduction on Copper in Aqueous Electrolyte. *Chem. Rev.* **2019**, *119*, 7610−7672.

(6) Andersen, S. Z.; Čolić, V.; Yang, S.; Schwalbe, J. A.; Nielander, A. C.; McEnaney, J. M.; Enemark-Rasmussen, K.; Baker, J. G.; Singh, A. R.; Rohr, B. A. A Rigorous Electrochemical Ammonia Synthesis Protocol with Quantitative Isotope Measurements. *Nature* **2019**, *570*, 504−508.

(7) Lu, X.; Li, Y.; Yang, P.; Wan, Y.; Wang, D.; Xu, H.; Liu, L.; Xiao, L.; Li, R.; Wang, G. Atomically Dispersed Fe-Nc Catalyst with Densely Exposed Fe-N_4 Active Sites for Enhanced Oxygen Reduction Reaction. *Chem. Eng. J.* **2024**, *485*, No. 149529.

(8) Dinh, C.-T.; Burdyny, T.; Kibria, M. G.; Seifitokaldani, A.; Gabardo, C. M.; García de Arquer, F. P.; Kiani, A.; Edwards, J. P.; De Luna, P.; Bushuyev, O. S. CO_2 Electroreduction to Ethylene Via Hydroxide-Mediated Copper Catalysis at an Abrupt Interface. *Science* **2018**, *360*, 783−787.

(9) Gasteiger, H. A.; Kocha, S. S.; Sompalli, B.; Wagner, F. T. Activity Benchmarks and Requirements for Pt, Pt-Alloy, and Non-Pt Oxygen Reduction Catalysts for Pemfcs. *Applied Catalysis B: Environmental* **2005**, *56*, 9−35.

(10) Wang, A.; Li, J.; Zhang, T. Heterogeneous Single-Atom Catalysis. *Nature Reviews. Chemistry* **2018**, *2*, 65−81.

(11) Kaiser, S. K.; Chen, Z.; Faust Akl, D.; Mitchell, S.; Pérez-Ramírez, J. Single-Atom Catalysts across the Periodic Table. *Chem. Rev.* **2020**, *120*, 11703−11809.

(12) Mohanty, B.; Jena, B. K.; Basu, S. Single Atom on the 2D Matrix: An Emerging Electrocatalyst for Energy Applications. *ACS omega* **2020**, *5*, 1287−1295.

(13) Chen, Y.; Ji, S.; Chen, C.; Peng, Q.; Wang, D.; Li, Y. Single-Atom Catalysts: Synthetic Strategies and Electrochemical Applications. *Joule* **2018**, *2*, 1242−1264.
(14) Shang, Y.; Xu, X.; Gao, B.; Wang, S.; Duan, X. Single-Atom Catalysis in Advanced Oxidation Processes for Environmental Remediation. *Chem. Soc. Rev.* **2021**, *50*, 5281−5322.
(15) Gusmão, R.; Veselý, M.; Sofer, Z. K. Recent Developments on the Single Atom Supported at 2D Materials Beyond Graphene as Catalysts. *ACS Catal.* **2020**, *10*, 9634−9648.
(16) Zhao, X.; Levell, Z. H.; Yu, S.; Liu, Y. Atomistic Understanding of Two-Dimensional Electrocatalysts from First Principles. *Chem. Rev.* **2022**, *122*, 10675−10709.
(17) Rebarchik, M.; Bhandari, S.; Kropp, T.; Mavrikakis, M. Insights into the Oxygen Evolution Reaction on Graphene-Based Single-Atom Catalysts from First-Principles-Informed Microkinetic Modeling. *ACS Catal.* **2023**, *13*, 5225−5235.
(18) Liang, W.; Chen, J.; Liu, Y.; Chen, S. Density-Functional-Theory Calculation Analysis of Active Sites for Four-Electron Reduction of O_2 on Fe/N-Doped Graphene. *ACS Catal.* **2014**, *4*, 4170−4177.
(19) Wan, X.; Liu, X.; Li, Y.; Yu, R.; Zheng, L.; Yan, W.; Wang, H.; Xu, M.; Shui, J. Fe−N−C Electrocatalyst with Dense Active Sites and Efficient Mass Transport for High-Performance Proton Exchange Membrane Fuel Cells. *Nature Catalysis* **2019**, *2*, 259−268.
(20) Jaouen, F.; Proietti, E.; Lefèvre, M.; Chenitz, R.; Dodelet, J.-P.; Wu, G.; Chung, H. T.; Johnston, C. M.; Zelenay, P. Recent Advances in Non-Precious Metal Catalysis for Oxygen-Reduction Reaction in Polymer Electrolyte Fuel Cells. *Energy Environ. Sci.* **2011**, *4*, 114−130.
(21) Kramm, U. I.; Herranz, J.; Larouche, N.; Arruda, T. M.; Lefèvre, M.; Jaouen, F.; Bogdanoff, P.; Fiechter, S.; Abs-Wurmbach, I.; Mukerjee, S. Structure of the Catalytic Sites in Fe/N/C-Catalysts for O_2-Reduction in Pem Fuel Cells. *Phys. Chem. Chem. Phys.* **2012**, *14*, 11673−11688.
(22) Zitolo, A.; Goellner, V.; Armel, V.; Sougrati, M.-T.; Mineva, T.; Stievano, L.; Fonda, E.; Jaouen, F. Identification of Catalytic Sites for Oxygen Reduction in Iron-and Nitrogen-Doped Graphene Materials. *Nature materials* **2015**, *14*, 937−942.
(23) Brimley, P.; Almajed, H.; Alsunni, Y.; Alherz, A. W.; Bare, Z. J.; Smith, W. A.; Musgrave, C. B. Electrochemical CO_2 Reduction over Metal-/Nitrogen-Doped Graphene Single-Atom Catalysts Modeled Using the Grand-Canonical Density Functional Theory. *ACS Catal.* **2022**, *12*, 10161−10171.
(24) Alsunni, Y. A.; Alherz, A. W.; Musgrave, C. B. Prediction of Potential-Dependent Kinetics for the Electrocatalytic Reduction of CO_2 to CO over Ti@4N-Gr. *ACS Electrochemistry* **2025**, *1*, 328−337.
(25) Zhao, Q.; Sun, J.; Li, S.; Huang, C.; Yao, W.; Chen, W.; Zeng, T.; Wu, Q.; Xu, Q. Single Nickel Atoms Anchored on Nitrogen-Doped Graphene as a Highly Active Cocatalyst for Photocatalytic H_2 Evolution. *ACS Catal.* **2018**, *8*, 11863−11874.
(26) Fung, V.; Hu, G.; Wu, Z.; Jiang, D.-e. Descriptors for Hydrogen Evolution on Single Atom Catalysts in Nitrogen-Doped Graphene. *J. Phys. Chem. C* **2020**, *124*, 19571−19578.
(27) Fei, H.; Dong, J.; Feng, Y.; Allen, C. S.; Wan, C.; Volosskiy, B.; Li, M.; Zhao, Z.; Wang, Y.; Sun, H. General Synthesis and Definitive Structural Identification of MN_4C_4 Single-Atom Catalysts with Tunable Electrocatalytic Activities. *Nat. Catal.* **2018**, *1*, 63−72.
(28) Cheng, N.; Stambula, S.; Wang, D.; Banis, M. N.; Liu, J.; Riese, A.; Xiao, B.; Li, R.; Sham, T.-K.; Liu, L.-M. Platinum Single-Atom and Cluster Catalysis of the Hydrogen Evolution Reaction. *Nat. Commun.* **2016**, *7*, No. 13638.
(29) Zhai, W.; Ma, Y.; Chen, D.; Ho, J. C.; Dai, Z.; Qu, Y. Recent Progress on the Long-Term Stability of Hydrogen Evolution Reaction Electrocatalysts. *InfoMat* **2022**, *4*, No. e12357.
(30) Zhang, C.; Wang, H.; Yu, H.; Yi, K.; Zhang, W.; Yuan, X.; Huang, J.; Deng, Y.; Zeng, G. Single-Atom Catalysts for Hydrogen Generation: Rational Design, Recent Advances, and Perspectives. *Adv. Energy Mater.* **2022**, *12*, No. 2200875.
(31) Ding, J.; Yang, H.; Zhang, S.; Liu, Q.; Cao, H.; Luo, J.; Liu, X. Advances in the Electrocatalytic Hydrogen Evolution Reaction by Metal Nanoclusters-Based Materials. *Small* **2022**, *18*, No. 2204524.
(32) Yang, Y.; Li, J.; Zhang, C.; Yang, Z.; Sun, P.; Liu, S.; Cao, Q. Theoretical Insights into Nitrogen-Doped Graphene-Supported Fe, Co, and Ni as Single-Atom Catalysts for CO_2 Reduction Reaction. *J. Phys. Chem. C* **2022**, *126*, 4338−4346.
(33) Zhang, Y.-F.; Yu, C.; Tan, X.-Y.; Cui, S.; Li, W.-B.; Qiu, J.-S. Recent Advances in Multilevel Nickel-Nitrogen-Carbon Catalysts for CO_2 electroreduction to CO. *New Carbon Materials* **2021**, *36*, 19−33.
(34) Hua, W.; Sun, H.; Lin, L.; Mu, Q.; Yang, B.; Su, Y.; Wu, H.; Lyu, F.; Zhong, J.; Deng, Z. A Hierarchical Single-Atom Ni-N3-C Catalyst for Electrochemical CO_2 Reduction to CO with near-Unity Faradaic Efficiency in a Broad Potential Range. *Chem. Eng. J.* **2022**, *446*, No. 137296.
(35) Pang, Q.; Fan, X.; Sun, K.; Xiang, K.; Li, B.; Zhao, S.; Kim, Y. D.; Liu, Q.; Liu, Z.; Peng, Z. Nickel−Nitrogen−Carbon (Ni−N−C) Electrocatalysts toward CO_2 Electroreduction to CO: Advances, Optimizations, Challenges, and Prospects. *Energy Environ. Mater.* **2024**, *7*, No. e12731.
(36) Wang, X.; Hu, Q.; Li, G.; Yang, H.; He, C. Recent Advances and Perspectives of Electrochemical CO_2 Reduction toward C_{2+} Products on Cu-Based Catalysts. *Electrochem. Energy Rev.* **2022**, *5*, 28.
(37) Ahmad, T.; Liu, S.; Sajid, M.; Li, K.; Ali, M.; Liu, L.; Chen, W. Electrochemical CO_2 Reduction to C_{2+} Products Using Cu-Based Electrocatalysts: A Review. *Nano Res. Energy* **2022**, *1*, No. e9120021.
(38) Tezak, C. R.; Singstock, N. R.; Alherz, A. W.; Vigil-Fowler, D.; Sutton, C. A.; Sundararaman, R.; Musgrave, C. B. Revised Nitrogen Reduction Scaling Relations from Potential-Dependent Modeling of Chemical and Electrochemical Steps. *ACS Catal.* **2023**, *13*, 12894−12903.
(39) Cui, X.; Tang, C.; Zhang, Q. A Review of Electrocatalytic Reduction of Dinitrogen to Ammonia under Ambient Conditions. *Adv. Energy Mater.* **2018**, *8*, No. 1800369.
(40) Guo, X.; Wan, X.; Shui, J. Molybdenum-Based Materials for Electrocatalytic Nitrogen Reduction Reaction. *Cell Rep. Phys. Sci.* **2021**, *2*, No. 100447.
(41) Liu, G.; Wang, J. Insights into Tuning of Mo-Based Structures toward Enhanced Electrocatalytic Performance of Nitrogen-to-Ammonia Conversion. *Adv. Energy Sustain. Res.* **2022**, *3*, No. 2100179.
(42) Nguyen, T. N.; Salehi, M.; Le, Q. V.; Seifitokaldani, A.; Dinh, C. T. Fundamentals of Electrochemical CO_2 Reduction on Single-Metal-Atom Catalysts. *ACS Catal.* **2020**, *10*, 10068−10095.
(43) Di Liberto, G.; Giordano, L.; Pacchioni, G. Predicting the Stability of Single-Atom Catalysts in Electrochemical Reactions. *ACS Catal.* **2024**, *14*, 45−55.
(44) Kim, G.; Lee, J.; Liu, T.; Grey, C. P. Characterizing Nitrogen Sites in Nitrogen-Doped Reduced Graphene Oxide: A Combined Solid-State 15N NMR, XPS, and DFT Approach. *J. Phys. Chem. C* **2021**, *125*, 10558−10564.
(45) Luo, E.; Chu, Y.; Liu, J.; Shi, Z.; Zhu, S.; Gong, L.; Ge, J.; Choi, C. H.; Liu, C.; Xing, W. Pyrolyzed $M-N_X$ Catalysts for Oxygen Reduction Reaction: Progress and Prospects. *Energy Environ. Sci.* **2021**, *14*, 2158−2185.
(46) Bates, J. S. Progress and Pitfalls in Designing Heterogeneous Catalysts with Molecular Precision. *Nat. Chem.* **2025**, *17*, 318−324.
(47) Lu, X.; Yang, P.; Wan, Y.; Zhang, H.; Xu, H.; Xiao, L.; Li, R.; Li, Y.; Zhang, J.; An, M. Active Site Engineering toward Atomically Dispersed M−N−C Catalysts for Oxygen Reduction Reaction. *Coord. Chem. Rev.* **2023**, *495*, No. 215400.
(48) Li, X.; Wang, D.; Zha, S.; Chu, Y.; Pan, L.; Wu, M.; Liu, C.; Wang, W.; Mitsuzaki, N.; Chen, Z. Active Sites Identification and Engineering of MNC Electrocatalysts toward Oxygen Reduction Reaction. *Int. J. Hydrogen Energy* **2024**, *51*, 1110−1127.
(49) Pan, Q.; Chen, Y.; Jiang, S.; Cui, X.; Ma, G.; Ma, T. Insight into the Active Sites of M−N−C Single-Atom Catalysts for Electrochemical CO_2 Reduction. *EnergyChem.* **2023**, *5*, No. 100114.
(50) Li, J.; Sougrati, M. T.; Zitolo, A.; Ablett, J. M.; Oğuz, I. C.; Mineva, T.; Matanovic, I.; Atanassov, P.; Huang, Y.; Zenyuk, I. Identification of Durable and Non-Durable FeN_X Sites in Fe−N−C Materials for Proton Exchange Membrane Fuel Cells. *Nat. Catal.* **2021**, *4*, 10−19.

(51) Vijay, S.; Gauthier, J. A.; Heenen, H. H.; Bukas, V. J.; Kristoffersen, H. H.; Chan, K. Dipole-Field Interactions Determine the CO_2 Reduction Activity of 2D Fe−N−C Single-Atom Catalysts. *ACS Catal.* **2020**, *10*, 7826−7835.
(52) Sundararaman, R.; Letchworth-Weaver, K.; Schwarz, K. A.; Gunceler, D.; Ozhabes, Y.; Arias, T. A. JDFTx: Software for Joint Density-Functional Theory. *SoftwareX* **2017**, *6*, 278−284.
(53) Sundararaman, R.; Goddard, W. A.; Arias, T. A. Grand Canonical Electronic Density-Functional Theory: Algorithms and Applications to Electrochemistry. *J. Chem. Phys.* **2017**, *146*, 114104.
(54) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Physical review letters* **1996**, *77*, 3865.
(55) Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, H. A Consistent and Accurate Ab Initio Parametrization of Density Functional Dispersion Correction (DFT-D) for the 94 Elements H-Pu. *J. Chem. Phys.* **2010**, *132*, 154104.
(56) Sundararaman, R.; Goddard, W. A. The Charge-Asymmetric Nonlocally Determined Local-Electric (CANDLE) Solvation Model. *J. Chem. Phys.* **2015**, *142*, No. 064107.
(57) Tezak, C.; Clary, J.; Gerits, S.; Quinton, J.; Rich, B.; Singstock, N.; Alherz, A.; Aubry, T.; Clark, S.; Hurst, R. Beast DB: Grand-Canonical Database of Electrocatalyst Properties. *J. Phys. Chem. C* **2024**, *128*, 20165−20176.
(58) Fromsejer, R.; Maribo-Mogensen, B.; Kontogeorgis, G. M.; Liang, X. Accurate Formation Enthalpies of Solids Using Reaction Networks. *npj Comput. Mater.* **2024**, *10*, 244.
(59) Zhang, X.; Wang, L.; Xie, Y.; Fu, H. Theoretical Insights into Performance Descriptors and Their Impact on Activity Optimization Strategies for Cobalt-Based Electrocatalysts. *Coord. Chem. Rev.* **2025**, *533*, No. 216560.
(60) Li, B.; Holby, E. F.; Wang, G. Mechanistic Insights into Metal, Nitrogen Doped Carbon Catalysts for Oxygen Reduction: Progress in Computational Modeling. *Journal of Materials Chemistry A* **2022**, *10*, 23959−23972.
(61) Huang, P.; Cheng, M.; Zhang, H.; Zuo, M.; Xiao, C.; Xie, Y. Single Mo Atom Realized Enhanced CO_2 Electro-Reduction into Formate on N-Doped Graphene. *Nano Energy* **2019**, *61*, 428−434.
(62) Zhang, C.; Sha, J.; Fei, H.; Liu, M.; Yazdi, S.; Zhang, J.; Zhong, Q.; Zou, X.; Zhao, N.; Yu, H. Single-Atomic Ruthenium Catalytic Sites on Nitrogen-Doped Graphene for Oxygen Reduction Reaction in Acidic Medium. *ACS Nano* **2017**, *11*, 6930−6941.
(63) Li, W.-X.; Stampfl, C.; Scheffler, M. Insights into the Function of Silver as an Oxidation Catalyst by Ab Initio Atomistic Thermodynamics. *Phys. Rev. B* **2003**, *68*, No. 165412.
(64) Huang, Y.; Chen, Y.; Xu, M.; Ly, A.; Gili, A.; Murphy, E.; Asset, T.; Liu, Y.; De Andrade, V.; Segre, C. U. Catalysts by Pyrolysis: Transforming Metal-Organic Frameworks (MOFs) Precursors into Metal-Nitrogen-Carbon (MNC) Materials. *Mater. Today* **2023**, *69*, 66−78.
(65) McQuarrie, D. A. *Quantum Chemistry*; University Science Books, 2008.
(66) Lai, L.; Potts, J. R.; Zhan, D.; Wang, L.; Poh, C. K.; Tang, C.; Gong, H.; Shen, Z.; Lin, J.; Ruoff, R. S. Exploration of the Active Center Structure of Nitrogen-Doped Graphene-Based Catalysts for Oxygen Reduction Reaction. *Energy Environ. Sci.* **2012**, *5*, 7936−7942.
(67) Menga, D.; Low, J. L.; Li, Y.-S.; Arčon, I.; Koyutürk, B.; Wagner, F.; Ruiz-Zepeda, F.; Gaberšček, M.; Paulus, B.; Fellinger, T.-P. Resolving the Dilemma of Fe−N−C Catalysts by the Selective Synthesis of Tetrapyrrolic Active Sites Via an Imprinting Strategy. *J. Am. Chem. Soc.* **2021**, *143*, 18010−18019.
(68) Auer, A.; Ding, X.; Bandarenka, A. S.; Kunze-Liebhauser, J. The Potential of Zero Charge and the Electrochemical Interface Structure of Cu(111) in Alkaline Solutions. *J. Phys. Chem. C* **2021**, *125*, 5020−5028.
(69) Huang, J.; Li, P.; Chen, S. Potential of Zero Charge and Surface Charging Relation of Metal-Solution Interphases from a Constant-Potential Jellium-Poisson-Boltzmann Model. *Phys. Rev. B* **2020**, *101*, No. 125422.
(70) Qin, N.; Yu, S.; Ji, Z.; Wang, Y.; Li, Y.; Gu, S.; Gan, Q.; Wang, Z.; Li, Z.; Luo, G. Oxidation State as a Descriptor in Oxygen Reduction Electrocatalysis. *CCS Chem.* **2022**, *4*, 3587−3598.
(71) Zhao, Z.; Schlexer Lamoureux, P.; Kulkarni, A.; Bajdich, M. Trends in Oxygen Electrocatalysis of 3 d-Layered (Oxy)(Hydro) Oxides. *ChemCatChem.* **2019**, *11*, 3423−3431.
(72) Tang, W.; Sanville, E.; Henkelman, G. A Grid-Based Bader Analysis Algorithm without Lattice Bias. *J. Phys.: Condens. Matter* **2009**, *21*, No. 084204.
(73) Yu, M.; Trinkle, D. R. Accurate and Efficient Algorithm for Bader Charge Integration. *J. Chem. Phys.* **2011**, *134*, No. 064111.
(74) Ertural, C.; Steinberg, S.; Dronskowski, R. Development of a Robust Tool to Extract Mulliken and Löwdin Charges from Plane Waves and Its Application to Solid-State Materials. *RSC Adv.* **2019**, *9*, 29821−29830.
(75) Huang, J.; Domínguez-Flores, F.; Melander, M. Variants of Surface Charges and Capacitances in Electrocatalysis: Insights from Density-Potential Functional Theory Embedded with an Implicit Chemisorption Model. *PRX Energy* **2024**, *3*, No. 043008.
(76) Huang, J.; Clinton, E. D.; Crossley, K.; Falqueto, J. B.; Schmidt, T. J.; Fabbri, E. Uncovering the Catalyst/Electrolyte Interfacial Process by Frequency Dispersion of Capacitance. *J. Energy Chem.* **2025**, *108*, 199−209.
(77) Binninger, T. First-Principles Theory of Electrochemical Capacitance. *Electrochim. Acta* **2023**, *444*, No. 142016.
(78) Schalenbach, M.; Selmert, V.; Kretzschmar, A.; Raijmakers, L.; Durmus, Y. E.; Tempel, H.; Eichel, R.-A. How Microstructures, Oxide Layers, and Charge Transfer Reactions Influence Double Layer Capacitances. Part 1: Impedance Spectroscopy and Cyclic Voltammetry to Estimate Electrochemically Active Surface Areas (ECSAs). *Phys. Chem. Chem. Phys.* **2024**, *26*, 14288−14304.
(79) Li, J.; Pham, P. H.; Zhou, W.; Pham, T. D.; Burke, P. J. Carbon-Nanotube−Electrolyte Interface: Quantum and Electric Double Layer Capacitance. *ACS Nano* **2018**, *12*, 9763−9774.
(80) Sánchez-Salas, R.; Kashina, S.; Galindo, R.; Cuentas-Gallegos, A. K.; Rayón-López, N.; Miranda-Hernandez, M.; Fuentes-Ramirez, R.; López-Urías, F.; Muñoz-Sandoval, E. Effect of Pyrrolic-N Defects on the Capacitance and Magnetization of Nitrogen-Doped Multiwalled Carbon Nanotubes. *Carbon* **2021**, *183*, 743−762.